WORKED SOLUTION

STEP III 2005, Question 1

Pure • Trigonometric Equations and Graphs

STEP III 2005, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q1

The key idea

Every equation of the form sin(something) = cos(something else) can be turned into a statement about the two *arguments*, using $\cos B = \sin(\frac{\pi}{2} - B)$. Applied to $\sin(\sin x) = \cos(\cos x)$, this says that $\sin x \mp \cos x$ would have to equal an odd multiple of $\frac{\pi}{2}$, and the bound on $|\sin x \pm \cos x|$ shows that it is simply too small to reach one. The sketches then follow from thinking about what the outer function does to an inner argument that only ranges over a short interval.

Solution

The equation $\sin A = \cos B$

Since $\cos B = \sin(\frac{\pi}{2} - B)$, we need the general solution of $\sin A = \sin C$ with $C = \frac{\pi}{2} - B$. The sine function takes each value in $[-1, 1]$ exactly twice in each period 2π, at C and at $\pi - C$ (these coincide when the value is ± 1), so

$$\sin A = \sin C \iff A = C + 2k\pi \quad \text{or} \quad A = \pi - C + 2k\pi \quad \text{for some integer } k.$$

Substituting $C = \frac{\pi}{2} - B$:

$$\begin{aligned} & A = \tfrac{\pi}{2} - B + 2k\pi = (4k+1)\tfrac{\pi}{2} - B, \\ \text{or} \quad & A = \pi - \tfrac{\pi}{2} + B + 2k\pi = (4k+1)\tfrac{\pi}{2} + B. \end{aligned}$$

So $\sin A = \cos B$ if and only if $A = (4n+1)\frac{\pi}{2} \pm B$ for some integer n. Both directions are covered, because each step above is an "if and only if". (For the "if" direction directly: $\sin((4n+1)\frac{\pi}{2} \pm B) = \sin(\frac{\pi}{2} \pm B) = \cos B$, using the period 2π.)

The bound and the equation $\sin(\sin x) = \cos(\cos x)$

Using the compound angle formula,

$$\sin x \pm \cos x = \sqrt{2}\left(\sin x \cos\tfrac{\pi}{4} \pm \cos x \sin\tfrac{\pi}{4}\right) = \sqrt{2}\sin(x \pm \tfrac{\pi}{4}),$$

and since $|\sin\theta| \le 1$ for every θ, we get $|\sin x \pm \cos x| \le \sqrt{2}$ for all x, as required.

Now suppose $\sin(\sin x) = \cos(\cos x)$ for some x. By the first result with $A = \sin x$ and $B = \cos x$, there is an integer n with

$$\sin x = (4n+1)\tfrac{\pi}{2} \pm \cos x, \qquad \text{that is,} \qquad \sin x \mp \cos x = (4n+1)\tfrac{\pi}{2}.$$

Ask yourself Why can't the right-hand side be small?

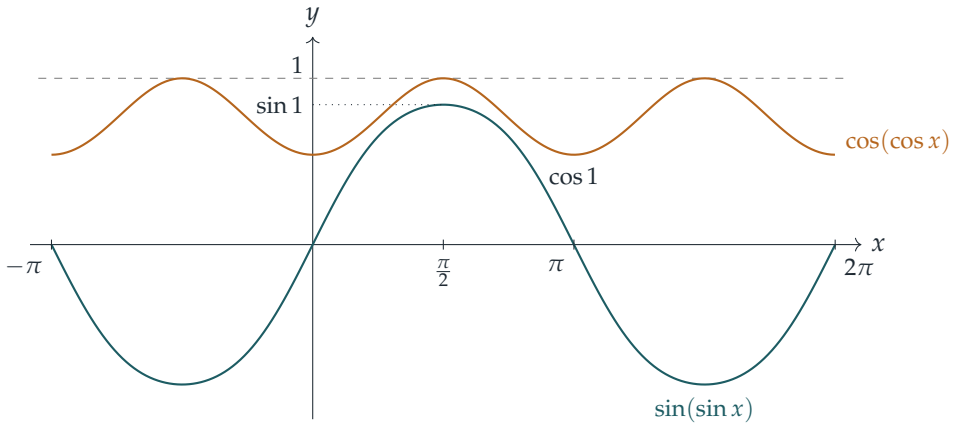
Because $4n+1$ is an odd integer, it is never 0, so $|4n+1| \ge 1$ and the right-hand side has modulus at least $\frac{\pi}{2} \approx 1.571$. The left-hand side has modulus at most $\sqrt{2} \approx 1.414$. Since $\sqrt{2} < \frac{\pi}{2}$ (square both: $2 < \frac{\pi^2}{4}$ because $\pi^2 > 9 > 8$), this is impossible. Hence $\sin(\sin x) = \cos(\cos x)$ has no solutions.

Sketching $y = \sin(\sin x)$ and $y = \cos(\cos x)$

$y = \sin(\sin x)$**.** The inner function $\sin x$ runs over $[-1, 1]$, and on that interval sin is increasing (as $[-1,1] \subset [-\frac{\pi}{2}, \frac{\pi}{2}]$). So $\sin(\sin x)$ rises and falls exactly when $\sin x$ does: it has period 2π, is odd, is zero at $x = n\pi$, has maximum $\sin 1 \approx 0.841$ at $x = \frac{\pi}{2} + 2n\pi$ and minimum $-\sin 1$ at $x = -\frac{\pi}{2} + 2n\pi$. Its gradient $\cos x \cos(\sin x)$ equals 1 at the origin. It looks like a slightly flattened sine curve.

$y = \cos(\cos x)$**.** Since $\cos(-u) = \cos u$ and $\cos(x+\pi) = -\cos x$, we have $\cos(\cos(x+\pi)) = \cos(\cos x)$: the graph has period π and is even. On $[-1,1]$, $\cos u$ is largest ($= 1$) at $u = 0$ and smallest ($= \cos 1 \approx 0.540$) at $u = \pm 1$. So $\cos(\cos x)$ has maxima 1 where $\cos x = 0$, i.e. at $x = \frac{\pi}{2} + n\pi$, and minima $\cos 1$ where $\cos x = \pm 1$, i.e. at $x = n\pi$. It always lies between 0.54 and 1.

By the previous part the curves never meet, and since $\cos(\cos 0) = \cos 1 > 0 = \sin(\sin 0)$, continuity tells us that the cosine curve lies strictly above the sine curve everywhere. They come closest (a gap of about 0.107) near $x \approx 0.88$, where $\sin(\sin x)$ is nearly at its peak and $\cos(\cos x)$ is still climbing.



Sketching $y = \sin(2\sin x)$

This again has period 2π and is odd, with zeros at $x = n\pi$, so it is enough to understand $0 \leq x \leq \pi$, where it is symmetric about $x = \frac{\pi}{2}$ (because $\sin(\pi - x) = \sin x$).

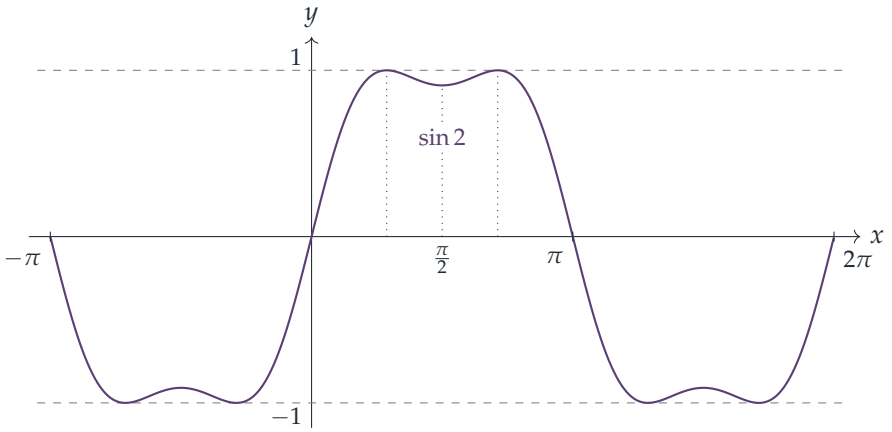
The difference now is that the inner argument $u = 2\sin x$ runs over $[0, 2]$ on this interval, and $2 > \frac{\pi}{2}$. So as x increases from 0 to $\frac{\pi}{2}$:

- u climbs from 0 to $\frac{\pi}{2}$ while $\sin x$ climbs to $\frac{\pi}{4}$, at $x = \arcsin\frac{\pi}{4} \approx 0.903$; meanwhile $y = \sin u$ rises from 0 to its maximum 1;
- u carries on from $\frac{\pi}{2}$ to 2, where $\sin u$ is decreasing, so y falls back to $\sin 2 \approx 0.909$ at $x = \frac{\pi}{2}$.

By symmetry, y rises again to 1 at $x = \pi - \arcsin\frac{\pi}{4} \approx 2.238$ and falls to 0 at $x = \pi$. Checking with the derivative,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\cos x \cos(2\sin x),$$

which is zero exactly when $\cos x = 0$ (the local minimum at $\frac{\pi}{2}$) or $2\sin x = \pm\frac{\pi}{2}$ (the maxima of 1); the gradient at the origin is 2. On $[\pi, 2\pi]$ the picture is reflected: two minima of -1 with a local maximum of $-\sin 2$ at $x = \frac{3\pi}{2}$ between them. Each hump of the sine curve has become a camel's back with a shallow dip.



Checks

- Take $B = 0.3$ and $n = 1$: $A = \frac{5\pi}{2} - 0.3$ gives $\sin A = \cos 0.3 \approx 0.9553$, as it should.
- Evaluating $\cos(\cos x) - \sin(\sin x)$ on a fine grid over a period gives a minimum of about 0.107 (near $x \approx 0.878$), never zero, which confirms that the curves don't meet.
- For $\sin(2\sin x)$, a numerical scan of $[0, \pi]$ finds exactly three turning points: maxima of 1 at $x \approx 0.903$ and 2.238, and a minimum of $\sin 2 \approx 0.909$ at $\frac{\pi}{2}$.

Common slips

- Writing only $A = \frac{\pi}{2} - B + 2n\pi$ and forgetting the second family of solutions $A = \frac{\pi}{2} + B + 2n\pi$, which comes from $\sin(\pi - C) = \sin C$.
- Forgetting that $n = 0$ is allowed and checking only $|4n + 1| \geq 5$: the crucial comparison is with $\frac{\pi}{2}$ itself.
- Drawing $\sin(2\sin x)$ as a plain sine curve of height 1: because $2 > \frac{\pi}{2}$, the inner argument overshoots the peak of sine, which causes the dip at $x = \frac{\pi}{2}$.

Beyond STEP

The sketches here are all about composition: the shape of $f(g(x))$ depends on which part of the graph of f the range of g covers. Iterating this idea ($\sin(\sin(\sin x))$, and so on) leads to fixed points and the dynamics of iterated maps, where one asks when $f(f(\cdots f(x)))$ settles down. The inequality argument, comparing the possible ranges of two sides, is a simple example of a technique used throughout analysis; Abbott's *Understanding Analysis* is a good place to see such arguments made rigorous.

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